

Total No. of Questions : 40]

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10thARFJD(S/Z) JKUT-2026

1303-A

MATHEMATICS

Time : 3 Hours]

[Maximum Marks : 80

SECTION-A

(1 mark each)

1. The L.C.M and H.C.F of 6 and 72 are :

(a) 6, 72

(b) 72, 6

(c) 12, 72

(d) None of these

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2. 12th term of the A.P. : 3, 1, -1, -3, is :

(a) 19

(b) 91

(c) -19

(d) None of these

3. If two lines are parallel, the system of equation has :

(a) One solution

(b) Two solutions

(c) No solution

(d) None of these

4. A cubic polynomial can have at most :

(a) 3 zeroes

(b) 2 zeroes

(c) 4 zeroes

(d) None of these

5. Sin 45° is equal to :
- (a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$
- (c) 2 (d) None of these
6. Which of the following is not a prime factor of 3825 ?
- (a) 3 (b) 17
- (c) 11 (d) None of these
7. The distance between the points (0, 3) and (4, 0) is :
- (a) 16 (b) 25
- (c) 5 (d) None of these

8. The discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$ is :

(a) 8

(b) 40

(c) ☒ -8

(d) None of these

9. The sum of first n natural numbers is :

(a) $\frac{n(n-1)}{2}$

(b) $\frac{n^2(n+1)}{2}$

(c) ☒ $\frac{n(n+1)}{2}$

(d) None of these

10. A card is drawn from a pack of 52 cards. The probability of drawing a face card is :

(a) $\frac{2}{13}$

(b) ☒ $\frac{3}{13}$

(c) $\frac{1}{13}$

(d) None of these

11. Any two similar figures are congruent.

(True / False)

12. A tangent to a circle intersects it in _____ point(s).

13. Define angle of depression.

Or

$\sin \theta = \cos \theta$ for all values of θ .

(True / False)

14. The sum of the exponents of prime factors in the prime factorisation of 250 is :

(a) 3

(b) 5

(c) 4

(d) None of these

15. The pair of equations $x = 2$ and $y = 3$ has :

(a) no solution (b) two solutions

(c) unique solution (d) None of these

16. The total surface area of a hemisphere of radius r , is :

(a) $2\pi r^2$ (b) $3\pi r^2$

(c) $\frac{2}{3}\pi r^3$ (d) None of these

17. The sum of the first 20 terms of the A.P. : 2, 7, 12, _____ is :

(a) 1040 (b) 1020

(c) 1000 (d) None of these

- Or**

The cumulative frequency of the last class is always equal to n .

Turn Over

SECTION-B

(2 marks each)

21. ✓ Find the zeroes of the quadratic polynomial $4s^2 - 4s + 1$.
22. ✓ Five years ago, Nuri was thrice as old as Sonu. Ten years later, Nuri will be twice as old as Sonu. How old are Nuri and Sonu?
23. ✗ Determine if the points (1, 5), (2, 3) and (-2, -11) are collinear.

Or

Find a relation between x and y such that the point (x, y) is equidistant from the points (3, 6) and (-3, 4).

24. ✓ Find the roots of the quadratic equation $2x^2 + x - 6 = 0$ by factorisation.

25✓ Given $\sec \theta = \frac{13}{12}$, calculate all other trigonometric ratios.

26✓ Find the slant height of a cone whose radius = 2.5 cm and height = 6cm.

Or

2 cubes each of volume 64cm^3 are joined end to end. Find the surface area of the resulting cuboid.

SECTION-C

(3 marks each)

27✓ A chord of a circle of radius 12cm subtend an angle of 120° at the centre. Find the area of the corresponding segment of the circle.

(use $\pi = 3.14$ and $\sqrt{3} = 1.73$)

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28. Find the ratio in which the line segment joining the points $(-3, 10)$ and $(6, -8)$ is divided by $(-1, 6)$.
29. Prove that opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.

Or

The length of a tangent from a point A at distance 5cm from the centre of the circle is 4cm. Find the radius of the circle. ✓



30. If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. Prove it. ✍

31. • A bag contains 3 red and 5 black balls. A ball is drawn at random from the bag. What is the probability that the ball drawn is

- (i) red (ii) not red

32. If the sum of first 7 terms of an A.P. is 49 and that of 17 terms is 289. Find the sum of first n terms.

Or

Find the 20th term from the last term of the A.P. : 3, 8, 13,,

253. •

33. Prove that $\sqrt{5}$ is irrational. •

34. S and T are points on the sides PQ and PR respectively of a ΔPQR . If $\frac{PS}{SQ} = \frac{PT}{TR}$ and $\angle PST = \angle PRQ$. Prove that PQR is an isosceles triangle. (

SECTION-D

(4 marks each)

35. Find two consecutive positive integers, sum of whose Squares is 365.

Or

Find the nature of the roots of the quadratic equation $2x^2 - 6x + 3 = 0$. If the real roots exists, find them.

36. E is a point on the side AD produced of a parallelogram ABCD and BE intersects CD at F. Show that $\triangle ABE \sim \triangle CFB$.

Or

A vertical pole of length 6m casts a shadow 4m long on the ground and at the same time a tower casts a shadow 28m long. Find the height of the tower. <https://www.jkboseonline.com>

37. Express the trigonometric ratios $\sin A$, $\sec A$ and $\tan A$ in terms of $\cot A$.

Or

- Prove the identity, where the angle involved is an acute angle

$$\frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} = 2 \sec A .$$

38. A Tower stands vertically on the ground. From a point on the ground, which is 15m away from the foot of the tower, the angle of elevation of the top of the tower is found to be 60° . Find the height of the tower.
39. A cubical block of side 7cm is surmounted by a hemisphere. What is the greatest diameter the hemisphere can have? Find the surface area of the solid.

Or

A solid toy is in the form of a hemisphere surmounted by a right circular cone. The height of the cone is 2cm and the diameter of the base is 4cm. Find the volume of the toy.

Turn Over

40. If the median of the distribution given below is 28.5, find the values of x and y .

Class interval	Frequency
0-10	5
10-20	x
20-30	20
30-40	15
40-50	y
50-60	5
Total	60
